3 SET THEORY

3.1 Introduction

Collection of unordered, district and well defined objects.

- {1, 2} = {2, 1} = {1, 2, 1} (order & repetition doesn't matter)

  Set representation: A {1, 2, 3} - Roaster form

  $A = \{z | z \in N \quad x \leq 3\}$ – Set builder form

- Set with one element is known as singleton set.
- Set with zero element is known as empty set or null set.
- No of elements in a set is known as cardinality of set.
- If a set A contains an element a, it is denoted as:

  $a \in A$.

3.1.1 Equal sets (Axiom of extension):

Two sets A and B are said to be equal iff they have same set of elements.

$A = B \Leftrightarrow \forall_x (x \in A \leftrightarrow x \in B) \Leftrightarrow A \subseteq B \wedge B \supseteq A$

3.1.2 Subset:

- A is said to be subset of B iff every element in A is also then in B. Denoted as $A \subseteq B$

  $\forall x (x \in A \Rightarrow x \in B)$

3.1.3 Proper subset:

- 'A' is said to be a proper subset of B if every element in A. exist in B and $A \neq B$. Denoted as $A \subset B$

  i.e., $A \subset B \Leftrightarrow \forall x (x \in A \rightarrow x \in B) \wedge \exists x. (x \in B \wedge x \notin A)$

  or

  $A \subset B \Leftrightarrow \forall x (x \in A \rightarrow x \in B) \wedge (|A| \neq |B|)$

3.1.4 Powerset:

Powerset of a set A is set of all subsets of A. Denoted as P(A)

- if $|A| = n$

  $|P(A)| = 2^n$

- For any set A

  $\phi \subseteq A$

  If $A = \phi$, then

  $\phi \subset A$

3.1.5 Operations on set:

(i) **Union:**

$A \cap B = \{x | x \in A \vee x \in B\}$

(ii) **Intersection:**

$A \cap B = \{x | x \in A \wedge x \in B\}$

(iii) **Complement:**

If A is a set,

Complement of A, $\bar{A} = A^c = \{x | x \in \cup \wedge \ x \notin A\}$

i.e., $\bar{A} = U - A = U \cap \bar{A}$

(iv) **Difference:**

$A/B = A - B = \{x | x \in A \wedge x \notin B\} = A \cap \bar{B}$

$A - \bar{B}$ is also called complement of B w.r.t. to A.

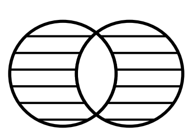
"complement of B in A"

(v) **Symmetric Difference:**

$A \Delta B = \{x | x \in A \vee x \in B$, but not both$\}$

Since it corresponds to XOR operator, it is also denoted as $A \oplus B$

$A \Delta B = (A - B) \cup (B - A) = (A \cup B) - (A \cap B)$



- Two sets are said to be disjoint if their intersection ϕ.

3.1.6 Loss involving set operations:

- $\left.\begin{matrix} A\cup A=A \\ A\cap A=A \end{matrix}\right\}$ Idempotent law
- $\left.\begin{matrix} A\cap\phi=\phi \\ A\cup U=U \end{matrix}\right\}$ Domination law
- $\left.\begin{matrix} A\cap U=A \\ A\cup\phi=A \end{matrix}\right\}$ Indentity law
- $\left.\begin{matrix} A\cup(B\cup C)=(A\cup B)\cup C \\ A\cup(B\cap C)=(A\cap B)\cap C \\ A\oplus(B\oplus C)=(A\oplus B)\oplus C \end{matrix}\right\}$ Associative law

  $\because$ XOR operation is associative
- $\left.\begin{matrix} A\cup(B\cup C)=(A\cup B)\cup(A\cup c) \\ A\cap(B\cap C)=(A\cap B)\cup(A\cap C) \end{matrix}\right\}$ Distributive law
- $\left.\begin{matrix} A\cup(A\cap B)=A \\ A\cap(A\cup B)=A \end{matrix}\right\}$ absorption law
- $\left.\begin{matrix} \overline{A\cup B}=\bar{A}\cup\bar{B} \\ \overline{A\cup B}=\bar{A}\cap\bar{B} \end{matrix}\right\}$ Demorgan's law

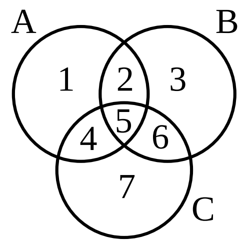
Note:

$\left.\begin{matrix} A\oplus B=A\oplus C\Rightarrow B=C \\ A\oplus C=B\oplus C\Rightarrow A=C \end{matrix}\right\}$ *The proot can be obtained by relating this to XOR operation in logic.*

3.1.7 Representation of sets using Venn diagrams:

Rather than direct calculations we sometimes represent sets with Venn diagram and see them as regions.

Let A, B, C be 3 sets



The region 1 is represented by $A\cap\bar{B}\cap\bar{c}$

The region 2 is represented by $A\cap B\cap\bar{c}$

Similarly, we have 8 regions including region outside of A, B, C

The principle of duality:

Let s denote a theorem dealing with the equality at two expressions which involve only operation $\cup$ and $\cap$. The dual of s(sd) (i.e., expression obtained by interchanging $\cup$ and $\cap$) is also a theorem.

Finding dual of $A \subseteq B$

$A \subseteq B$ can be written as

$A\cup B = B$

Dual is $A\cap B = B$

i.e., $B \subseteq A$

$\therefore$ Dual of $A \subseteq B$ is $B \subseteq A$

Duality principle should be applied only for general can but no for particular cases

3.1.8 Cartesian Product:

$A \times B = \{(a, b) | a \in A \wedge be B\}$

- Cartesian product is associative

  However

$$\begin{array}{ccc} (A\times B)\times C & \neq & A\times(B\times C) \\ \downarrow & & \downarrow \\ \textit{it has ordered pairs t type}\ ((a,b),c) & & (a,(b,c)) \end{array}$$

  $A \times B \neq B \times A$

  If $A \times B = B \times A$ then $(A = B \vee (A = \phi) \vee (B = \phi)$

- $A \times (B\cap C) = (A \times B) \cap (A \times C)$

  $A \times (B\cup C) = (A \times B) \cup (A \times C)$

  $\bar{A}\times\bar{B} \neq \overline{A\times B}$